CHANDU PRASAD BHAIRAPU



AI Engineer

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github

ABOUT ME

Master's graduate in Artificial Intelligence with 3 years of professional experience in AI engineering and software development. Specialized in LLM applications and agents, from prompts, context and agent workflows to evaluations and cloud deployment, built end-to-end from interface to backend. Most recently sole developer of an agentic knowledge-extraction system for the German energy sector, backed by AI-assisted development with Claude Code and stakeholder consulting across enterprise and start-up environments.

EXPERIENCE

Associate Generative AI Developer - Agentic Indexing Pipeline Engineer for German Energy Sector

Kniit GmbH

November 2025 – present Muldestausee, Germany

- Owned the knowledge-extraction pipeline end-to-end as sole developer, from data ingestion and chunking to prompt engineering and indexing (agentic, deterministic and monolithic strategies), reporting directly to the CTO and working with SAP domain experts across the German energy sector.
- Designed and optimized the multi-agent orchestration in LangGraph (entity-extraction, relationship-analysis, consolidation and reflection agents running in parallel) for complex regulatory and technical documentation, raising precision/recall from 33%/45% to 93%/90% measured against an SAP-system-derived ground truth.
- Engineered a deterministic, code-based extraction pipeline for a second, high-volume source stream, cutting a $200,000 projected LLM budget to near-zero recurring cost while pushing accuracy toward 100%.
- Designed and implemented a Model Context Protocol (MCP) service that exposes the knowledge graph for millisecond-latency retrieval, integrated into Claude Code for agentic development workflows.
- Developed Expert Interface, a graph curation web app (React frontend, Python microservices backend) on Azure where domain experts validate and approve knowledge-graph changes.
- **Tech stack: Python, LangGraph, LangChain, LiteLLM, MCP, Claude Code, Neo4j, GraphRAG, RAG, Vector Search, FastAPI, MongoDB, Postgres, Docker, Kubernetes, Azure, React**

GenAI Developer (AI Consulting Intern) - Development and Deployment of an Agentic AI Workflow

Covestro Deutschland AG

December 2024 – April 2025 Leverkusen, Germany

- Worked with business stakeholders to understand their decision-making workflows, advised on where agentic AI could support them and translated their requirements into the solution approach.
- Designed and deployed an agentic AI application with LangGraph that routes decision-making across vector database queries, general inference and summarization.
- Built the Python/FastAPI backend and a React frontend, tested the APIs with Postman, containerized the stack with Docker and deployed it to AWS with CDK; versioned in GitLab with CI/CD pipelines.
- **Tech stack: Python, LangGraph, LangChain, FastAPI, Docker, AWS (CDK, EC2, Lambda), GitLab CI/CD, Postman, React**

GenAI Developer (Master's Thesis) - Evaluating the Performance of RAG Systems

Covestro Deutschland AG

May 2024 – November 2024 Leverkusen, Germany

- Automated evals of RAG generation and retrieval with RAGAS and PytrecEval, against chemist-provided ground truth, with serverless AWS infrastructure for PrismaChat RAG evaluation (Lambda, SQS, DynamoDB) built with CDK.
- Benchmarked multilingual embedding models on over 1M chunks of internal chemical data and designed RAG pipelines (chunking, vector databases, indexing, fine-tuning models for retrieval).
- **Tech stack: Python, RAGAS, PytrecEval, LangChain, Hugging Face Transformers, PyTorch, AWS (Lambda, SQS, DynamoDB, CDK), Azure AI Search, Docker**

Data Scientist (Intern) – ML-Based Prediction of Prepolymer Synthesis Outcomes

Covestro Deutschland AG

November 2023 – April 2024 | Leverkusen, Germany

- Built and compared Random Forest, XGBoost and LSTM models to predict prepolymer synthesis outcomes from time-series process data, and delivered a Streamlit dashboard for chemists to simulate parameter changes before running physical batches.
- **Tech stack: Python, Scikit-Learn, XGBoost, TensorFlow/Keras, Pandas, NumPy, Streamlit**

Machine Learning Engineer - Electrical Vehicle Load Forecasting using Time Series Analysis

Cognizant Technology Solutions

September 2021 – March 2022 | Hyderabad, India

- Developed ARIMA, SARIMA, XGBoost regressor, FB Prophet and LSTM models for EV load time-series forecasting; the LSTM reached 96% test accuracy with strong performance on unseen data.

PROJECTS

Project: Knowledge Assistant for Internal Documents | LlamaIndex | MCP | REST API

- RAG assistant over internal process documents and intranet pages, built with LlamaIndex and a locally hosted open-source LLM; answers cite their sources and are served through a FastAPI REST API and an MCP server, so AI agents can call it as a tool.
- pytest unit tests and a RAGAS evaluation set run in a GitLab CI/CD pipeline; request logging with latency and retrieval metrics for monitoring; MkDocs documentation of architecture, API, and operation.

Project: Agent Eval Harness | Python | LangGraph | pytest | OpenTelemetry | GitHub Actions

- Evaluation harness for a tool-calling LangGraph agent over a mocked API: golden datasets (regression, edge cases, synthetic multi-turn, adversarial), deterministic checks first (JSON schema, function-calling payloads, regex, API status, embedding similarity), then LLM-as-a-judge only for what they cannot decide.
- Runs each case several times and reports pass rates with confidence intervals; tests prompt injection, malformed inputs, unexpected tool responses and tool-loop limits; LLM response caching and tool mocking in GitHub Actions, tracing and observability with OpenTelemetry and Arize Phoenix.

TECHNICAL SKILLSET

- **LLMs & Agents:** LangGraph agent workflows, LangChain, LiteLLM, MCP tools, evaluations against ground truth (precision/recall, RAGAS, PytrecEval), embedding model benchmarking, deterministic code-based extraction, prompt engineering, RAG, GraphRAG
- **AI-Assisted Development:** coding agents, Claude Code (custom skills, MCP integration, agentic coding workflows)
- **Programming & Backend:** Python, FastAPI, Python microservices, SQL, Postman, React (beginner)
- **Data & Knowledge:** Neo4j knowledge graphs, MongoDB, Postgres, vector databases (Qdrant, Pinecone, Azure AI Search), Hugging Face Transformers, PyTorch, Scikit-Learn
- **Cloud & DevOps:** Docker, Kubernetes, Azure, AWS (EC2, Lambda, SQS, DynamoDB, S3, CDK), Git, GitLab CI/CD

EDUCATION

Master of Science in Artificial Intelligence — March 2022 – April 2025
Technische Hochschule Würzburg-Schweinfurt, Würzburg, Germany

Bachelor of Technology in Mechanical Engineering — August 2017 – July 2021
Minor in Computer Science & Engineering
G Pulla Reddy Engineering College, Kurnool, India

PUBLICATIONS & CERTIFICATIONS

- **[1]** Navigating the Future: An approach of Autonomous Indoor Vehicles
- Udemy - LLMs Mastery : Transformers & Generative AI

LANGUAGES

English (Business Fluent) | German (B1) | Hindi (Fluent) | Telugu (Native)